

3(a)	$(m =) \rho V$	C1
	$= 1.0 \times 10^3 \times 1.5 \times 10^{-4} \times 5.0 \times 1.6 = 1.2 \text{ (kg)}$	A1
3(b)(i)	$(\Delta)p = 1.2 \times 5.0$ $= 6.0 \text{ N s}$	A1
3(b)(ii)	$F = 6.0 / 1.6$ or $1.2 \times 5.0 / 1.6$ $= 3.8 \text{ N}$	A1
3(c)	Newton's third law applies (so) 3.8 N.	B1
3(d)	$p = F / A$ $= 3.8 / 1.5 \times 10^{-4}$	C1
	$= 2.5 \times 10^4 \text{ Pa}$	A1